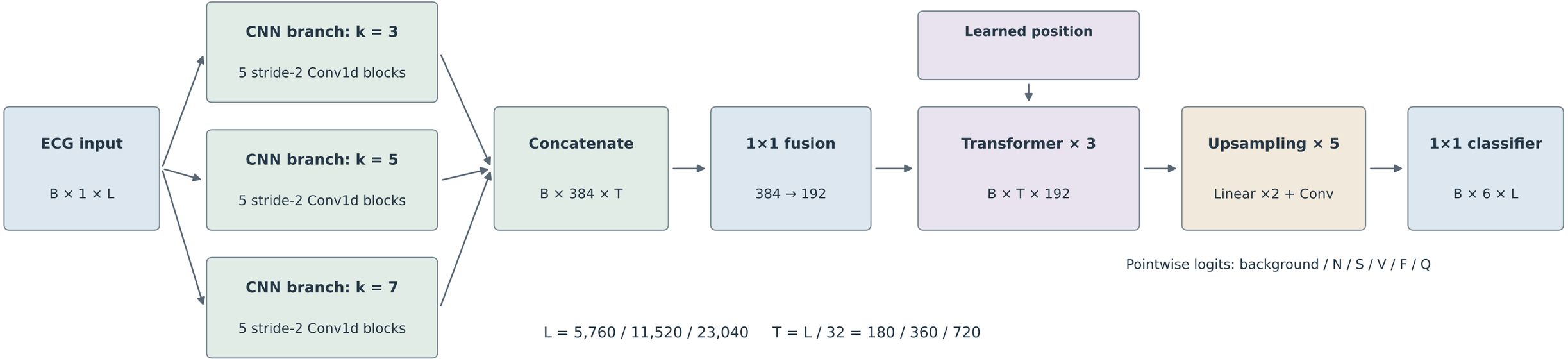
(a) Multi-scale CNN–Transformer for dense heartbeat classification



(b) CNN branch

Conv1d → GroupNorm(8) → GELU

kernel k; stride 2; padding floor(k/2)

Channels: 1 → 8 → 16 → 32 → 64 → 128

The same kernel size is used in all five layers.

(c) Transformer encoder block

LayerNorm → Self-attention → Add

6 heads; residual connection

LayerNorm → Feed-forward → Add

192 → 384 → 192; GELU; dropout 0.1

Non-causal attention over the entire input window

(d) Dense decoder

Linear upsampling ×2 → Conv1d(k=3)

GroupNorm(8) → GELU; repeated five times

Channels: 192 → 128 → 96 → 64 → 32 → 32

Final 1×1 convolution: 32 → 6 class logits